Performance Evaluation of Data Center Network with Network Micro-segmentation

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***Abstract*—Research on the design of data center infrastructure is increasing, both from academia and industry, due to the rapid development of cloud-based applications such as search engines, social networks, and large-scale computing. On a large scale, data centers can consist of hundreds to thousands of servers that require systems with high-performance requirements and low downtime. To meet the network's needs in a dynamic data center, infrastructure of applications and services are growing. It takes a process of designing a network topology so that it can guarantee availability and security. One way to surmount this is by implementing the zero trust security model based on micro-segmentation. Zero trust is a security idea based on the principle of "never trust, always verify" in which no concepts of trust and untrust in network traffic. The zero trust security model implemented network traffic in the form of untrust. Microsegmentation is a way to achieve zero trust by dividing a network into smaller logical segments to restrict the traffic. In this research, data center network performance based on software-defined networking with zero trust security model using micro-segmentation has been evaluated using a testbed simulation of Cisco Application Centric Infrastructure by measuring the round trip time, jitter, and packet loss during experiments. Performance evaluation results show that micro-segmentation adds an average round trip time of 4 µs and jitter of 11 µs without packet loss so that the security can be improved without significantly affecting network performance on the data center.**

Keywords—data center network, micro-segmentation, software defined networking, zero trust security

I. INTRODUCTION

Data center network requirements can be met using interconnected switches, which is referred to as data center networks in this paper [1]. In its application, most of the data center network technology uses ethernet technology. The use of Ethernet technology on data center networks is implemented using the Ethernet-switched or Internet Protocol-routed (IP- routed) method. Ethernet-switched and IP-routed, each has advantages and disadvantages; one of the essential factors to consider is one a high level of flexibility in development. Flexibility aims to meet the availability of networks for interconnection from thousands to millions of servers if one day is needed on a data center network. Somethings that need to be considered in technology selection are the ability to handle massive traffic (packet forwarding), routing features efficiently, and the ability to process migration on a virtual machine (VM) [2].

Problem-solving in data center networks using approaches by following IETF standards is an interesting topic. Greenberg et al. [3] and Zhang et al. [4] explain the problems that are often encountered in data center networks, among others, in terms of flexibility of development (scalability), data integrity, throughput, and the level of availability. The flexibility of development (scalability), in addition to the flexibility in installing a large number of servers and the flexibility in progress are essential factors in the future. Another problem with data center network systems is the use of static resources using VLAN mechanisms based on each application's needs. In addition to the advantages of VLANs in terms of performance and network traffic isolation, VLANs is lacking on the agility. This is because the concept of VLAN integrates many aspects such as network traffic management, security and isolation performance, and traffic on a VLAN becomes one, causing a high level of oversubscription on core networks. In conventional data center networks, the design of load balancing mechanisms is limited by the applications deployed. The fragmentation of resources sorted out this problem by dividing them into smaller subnets to restrict access to resources that are not needed at the same time. We conducted simulations to determine micro-segmentation performance by measuring round trip time, jitter, and packet loss on east-west data center traffic using a testbed simulation. This research was conducted on east-west traffic using SDN technology that focuses on data centers that makes the network more flexible, dynamic, and cost-effective. SDN can simplify the complexity of operations, thereby increasing performance on network systems [5]. The testbed simulation using a Cisco Application Centric Infrastructure (ACI) is based on software-defined networking with zero trust security model based on micro-segmentation because of programmable fabrics, dynamically provisioned, and scalable, Cisco ACI is the best choice [6].

The structure of the rest of this paper is as follows. We provide the related work in Section 2. In Section 3, we show the material and method. Section 4 presents the results and discussion for the evaluation performance of micro-segmentation simulation in detail and provides evaluation performance studies. In Section 5, we offer a conclusion and future work.

II. RELATED WORKS

Hernandez, L. et al. [7] conducted research using the methodology of preparation, planning, design, implementation, operation, and optimization (PPDIOO) by building and comparing conventional IP network simulation systems and Software-Defined Networking (SDN). Traditional IP network simulations run on Enhanced Interior Gateway Routing Protocol (EIGRP) and Border Gateway Protocol (BGP) protocols using GNS3 simulators. In contrast, SDN-based network simulation systems with OpenFlow protocols and POX controllers use Mininet simulators. In this study, the variables on network performance measured and compared are delay, throughput, and jitter. The test was carried out using a ping mechanism

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using IPv4 and IPv6 as many as 500 data packets with variable-size data sent by 200 Bytes, 1,000 Bytes, 10,000 Bytes, and 15,000 Bytes. In the process of measurement with the ping mechanism, traffic recording is recorded on the network using the Wireshark tool, the results used as analysis material. The study results showed that the measurement results using an SDN-based network system showed better performance than using a traditional IP-based network system both in measurements using IPv4 and IPv6. On the other hand, the SDN network system is a new technology in network design and implementation that is dynamic, flexible, and easy to implement for current network and application needs.

Kasi, A. A., et al. [8] conducted a study using the testbed simulation method by analyzing the core switch network's performance by building a testbed simulation using a switch device with different brands and types, namely 3Com 3500 and Cisco 3550. The study was conducted by developing a testbed simulation system with five topology systems. The difference is topology without VLAN, topology with the same VLAN (intra-VLAN), topology with different VLANs (inter-VLAN), topology with different VLANs and operating systems, and topologies with spanning tree protocols. The variables used to measure performance are consistency, throughput, and latency. Performance measurement is done by sending a data packet of 589 MB. As many as eight times, the transmission is made between 2 computer systems using the FileZilla tool and then recording the traffic using the Wireshark tool for the analysis process. In addition to this, measurements were made using the ping mechanism and recording using Wireshark to determine the topology's performance with the spanning tree protocol. The results of performance measurements using a testbed simulation with five scenarios show that 3Com shows a consistent consistency but is lower than Cisco, which shows inconsistent consistency. Cisco shows better data transmission compared to 3Com. On the other hand, 3Com shows better performance in sending data at the Layer 2 mechanism. Based on the results of performance measurements show that Cisco has better performance when it functions as a distribution and core layer (L3). In comparison, 3Com has better performance can be operated as access (L2).

Vuletic, D. V. et al. [9] conducted research to measure computer systems' performance against distributed denial of services (DDoS) attacks. The study was carried out by building the same network system between the victim and the Windows operating system and the attacker with the Linux operating system (Kali Linux). Testing is done by running the ping mechanism and carrying out DDoS attacks using the hping3 tool from Kali Linux. Hping3 is a free tool that can be used to simulate network traffic in the form of TCP / IP protocol by generating network traffic according to needs such as TCP Sync flood to flood network channels so that the availability of network systems is disrupted. The results of testing using hping3 show that DDoS attacks succeeded in disrupting service performance for victims. The result is indicated in the CPU process has increased. On the other hand, monitoring attack suggests that the ping process means a request time out so that the service becomes unavailable.

Surantha, N. et al. [10] conducted a study to test the performance of network system security design using planning, design, implementation, operation, and optimization (PDIOO) and building a testbed simulation environment in the form of a cubicle container system design by implementing network security using a zero trust model. Tests on the testbed simulation environment in the way of propagation of data transmission paths in network traffic (routing) using the BGP protocol and port scanning using the Nmap tool. The test is carried out using the concept of zero trust in which all network traffic is considered untrustworthy by applying a configuration that rejects all network traffic. The test results show that the testbed simulation's design using the zero trust security model was successfully implemented by preventing attacks in the form of route hijacking and scanning on the network from external parties by applying a configuration on the perimeter firewall. On the other hand, the use of labels on Kubernetes is used to limit network traffic to the internal container environment by restricting communication to unidentified and unauthorized network traffic, looking at the tags used on the test mechanism.

III. The Material and Method

A. *Software Defined Networking*

Software-Defined Networking (SDN) is a relatively new technology for designing and controlling network devices, so that limitations on network devices can be overcome, using two methods [11]. First, the separation between the control plane and the data plane; in this case, the network device functions only to forward the data packet so that it becomes simpler and improves performance. Second, the division of vertical integration functions in logical systems that control the network with network devices that forward data packets. Logical tasks that control the network can be implemented using a centralized controller to facilitate the configuration and application of policies relating to the system. Besides, the separation of the control plane with the data plane aims to promote the development of innovation by adding other features on the network such as load balancing, firewalls, IDS/IPS, and others.

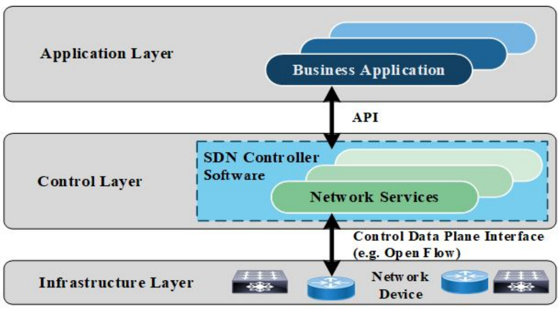

Fig. 1. Architecture of SDN [12]

Fig. 1 shows the architecture of SDN. According to S. Sezer et al. [12], there are four ways to identify SDN technology. First, there is a separation in the control plane and the data plane, so that the device functions only forward the data packet. In contrast, the device control process is done separately using a controller. Second, monitoring all components of network devices can be done using a centralized controller. Third, the protocol used to communicate between the controller and the device. Fourth, network programming can be done using external applications (third-party).

B. The Zero Trust Security Model

The zero trust security model is a security concept that is based on the principle of never trust, always verify. The core concept of zero trust is that there are no more trusted, untrusted user interfaces, networks, and users in the security device.

Fig. 2. All interface are untrust [13]

Fig. 2 shows that zero trust treats all network traffic as untrusted. Zero trust does not say that employees cannot be trusted, but trust is a concept that should not be applied by information security. The zero trust security model's basic idea is to verify and secure all resources, tightly restrict access, control, check, and record all network traffic [13]. First, the application of security and verification of resources is carried out on all access regardless of the source originating. All traffic is considered a threat and is not trusted so that traffic is authorized, checked, and secured. The application of internal security is carried out as the case with external protection. Second, access restrictions are strictly implemented by applying access controls correctly by the rules made so that access to resources is limited.

C. The Advantage of Zero Trust Security Model

The conventional security model uses the concept of trust and untrust in network traffic, from now on, referred to as the perimeter concept [14]. In building conventional security models, an organization's concept of the perimeter has been used to facilitate access from outsiders so that the mitigation process against attacks can run optimally. In the perimeter concept, the capability is needed in securing using security devices such as firewalls and VPNs. The need for security increases with the development of technologies such as cloud computing and the internet of things that cause cyber attacks, making the perimeter concept did not run optimally in preventing cyber attacks. This concept is not worth the investment that has been spent to build the idea of the perimeter. At present, the need for a security system is essential to improve organizational security [13], with a zero trust security model [14].

Implementing a zero trust security model can provide a high-security system, so it has a positive impact on business processes, such as the company's visibility in protecting customer data. The adoption of a zero trust security model can provide benefits from the organization's side to external parties with a good reputation in implementing security systems and budget savings in security system audits. On the inner side of the company, it has high feasibility and use of time to be efficient in monitoring security systems with high visibility, thereby reducing the complexity of implementing security systems. Service needs and easy maintenance of security systems are other benefits that can be obtained by applying zero trust [15]. The zero trust security model cannot be 100% fully capable but can mitigate well and reduce the occurrence of cyber-attacks [16].

Cisco's zero trust is a thorough approach to securing all access by networks, applications, and environments. One of the zero trust security models at Cisco is zero trust for workloads, which minimizes lateral movements that comprise breaches with micro-segmentation using Cisco ACI [17].

D. Micro-segmentation

Micro-segmentation is a way of dividing networks into smaller segments to limit network traffic. Micro-segmentation has an advantage over traditional perimeter security systems in that a smaller section can reduce the broader damage from malicious software (malware) attacks, and access to resources can be controlled based on established rules [18]. Micro-segmentation is an implementation of a distributed firewall that regulates access to network traffic based on security rules that have been determined on each resource. At present, both Cisco ACI and VMWare NSX are leading vendors in micro-segmentation, each of which has advantages and disadvantages [19]. Fig. 3 shows the concept of micro-segmentation in east-west network traffic in the data center.

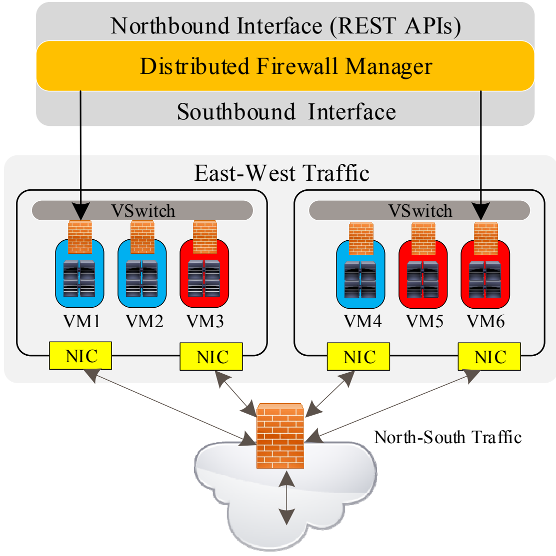

Fig. 3. Micro-segmentation concept [19]

Along with the development of software-defined networking (SDN) and software-defined data center (SDDC), micro-segmentation is now an increasingly popular technology because it has a high level of security and flexibility in its implementation. Micro-segmentation facilitates work in network administration and enforcement of security rules in data centers. With micro-segmentation, smaller segments can limit access rights on users, applications, and servers. Thereby it reduced the more extensive damage from malware attacks. Besides, access to resources can be controlled based on rules that have been made [19].

E. Cisco Application Centric Infrastructure (ACI)

Cisco Application Centric Infrastructure (ACI) is a concept in software-defined networking (SDN) by implementing architecture under application requirements. That way, that network implementation can be adjusted based on application requirements. This architecture aims to simplify, optimize, and accelerate an application's development cycle. The Cisco Application Policy Infrastructure Controller (APIC) is an application programming interface (API) used to connect applications, the network, and storage securely. APIC makes it easy for network administrators to define the optimal network for an application. APIC provides management and automation, use of policies by programming, application implementation, monitoring the condition of the fabric, and regulating the scalability of various tenants in the fabric [20].

Cisco ACI fabric consists of Cisco Nexus 9000 series and APIC built using ACI leaf/spine fabric by adopting a fat-tree network topology that connects each leaf switch to the spine switch controlled by APIC. In contrast, all devices are connected to the leaf switch. The recommended configuration on APIC uses a minimum of 3 servers in groups. ACI fabric can send traffic with low lag times because it runs on networks with a bandwidth capacity of 40 Gigabit per second (Gbps). Network traffic from the source and destination on the same leaf will be carried out locally, while traffic from the source and destination on the different leaf will pass through the spine first. The ACI fabric architecture runs on L3 even though there are only two hops physically [20].

End Point Group (EPG) is an object that can be controlled and is called a logical entity that contains a set of endpoints (endpoints). Endpoints are resources that are connected to the network both directly and indirectly and have addressed (identity), location, attributes, and can be either physical or virtual resources. EPG is wholly separated from physical and logical topology. Servers, virtual machines, and network-attached storage are examples of endpoints that can be dynamic or static. Micro-segmentation in Cisco ACI is the interconnecting endpoints of several EPGs into micro-segmented EPGs based on machine attributes, IP address, or MAC address. Virtual machines can have vNIC domain name attributes, VM markers, VM names, hypervisor markers, VMM domains, datacenter, operating systems, or self-defined attributes [20].

F. Methods and Design Scenario

We conducted this research using the Preparation, Planning, Design, Implementation, Operation, and Optimization (PPDIOO) methodology.

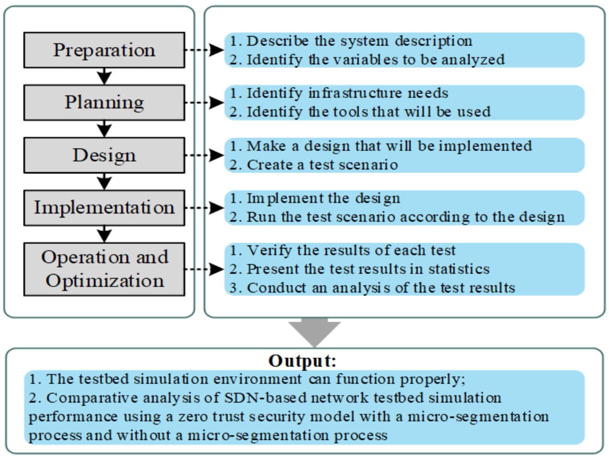

Fig. 4. PPDIOO Methodology

Fig. 4 shows the PPDIO methodology from Cisco [21] that defines the cycle in building a network system with sustainable services for providing a high level of availability. Fig. 4 shows the process's stages to be carried out to produce an output target by the research objectives.

A testbed simulation system to test performance was built using an SDN-based network system based on a micro-segmentation-based zero trust security model. The testbed simulation infrastructure built consisted of network infrastructure using Cisco ACI as an illustration of a data center network system with a research focus on east-west traffic. While the server infrastructure as an illustration of the application system services consisting of 3 tier applications, namely web server (web), app server (app), and database server (DB), by using a virtual machine (VM). The test scenario is carried out by measuring network performance in the form of round trip time, jitter, and packet loss with and without a micro-segmentation process, a feature that a network system has on a Cisco ACI against 3 tier web systems, apps, and DB. Fig. 5 shows the testbed simulation topology.

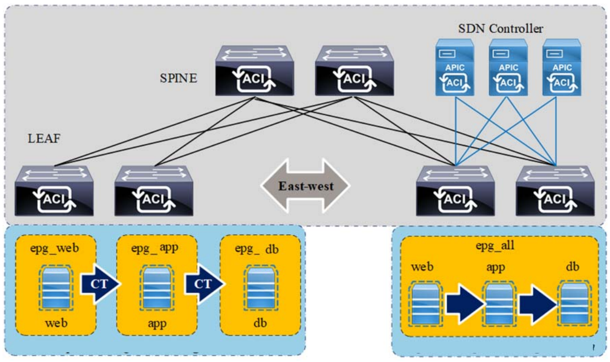

Fig. 5. The testbed simulation topology

Testing on network performance can be done using many variables. In this study, testing focused on three variables: round trip time (RTT), jitter, and packet loss with and without micro-segmentation processes on 3 tier web, app, and DB systems. Round Trip Time (RTT) is a unit of time required to send a packet the total time needed to initiate the interconnection process and transmit data packets so that the RTT is a delay time consisting of propagation time between 2 hosts in a network system. RTT in a network system is known as ping time. Jitter is a variety of delays that occur in consecutive nodes whose value depends on the burden of data traffic and the amount of collision between packets in a network system. The more significant weight of data traffic is a factor that affects the chances of a more substantial collision, changing the higher value of jitter. Packet loss is the occurrence of failed data packet delivery to achieve its destination.

The scenario of network performance testing is done by sending data packets of 100 packets with significant data variations of 200 bytes, 500 bytes, 1,000 bytes, 10,000 bytes, and 15,000 bytes. Data packet delivery is carried out from the web to the app and app to the DB by using the ping mechanism. The data traffic is recorded using Wireshark for analysis of Round Trip Time (RTT), jitter, and packet loss. The results of network performance testing are carried out with a micro-segmentation process. Without a micro-segmentation process using a simulated ACI fabric testbed, the results of both are analyzed. Network performance testing is done using a test tool in the form of Wireshark to obtain Round Trip Time (RTT), Jitter, and Packet Loss information. Wireshark is a network packet analyzer that can function to detect and display packets that pass through a network in detail.

Fig. 6 shows the testbed simulation design without the micro-segmentation process. We conduct the testbed simulation design without a micro-segmentation on the ACI fabric with the following steps. First, creating EPG, i.e., Web, App, and DB, are in the same EPG as epg_all. Second, configuring the BD and EPG epg_all with the aim of the Web VM, App, and DB entering the network segment /24.

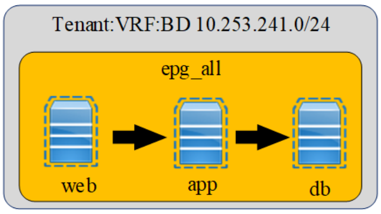

Fig. 6. Experiment without Micro-Segmentation Process

Fig. 7 shows the testbed simulation design with the micro-segmentation process. We conduct the testbed simulation design with a micro-segmentation on the ACI fabric with the following steps. First, creating EPG, i.e., web, app, and DB, is in a different EPG with epg_web, epg_app, and epg_db. Second, configuring the BD and EPG epg_web, epg_app, epg_db with the destination VM web, app, and DB entering the network segment /24. Third, creating a contract (CT), a policy on the network that regulates the interconnection between epg_web to epg_app, epg_app to epg_db with access policy in the form of ICMP (ping).

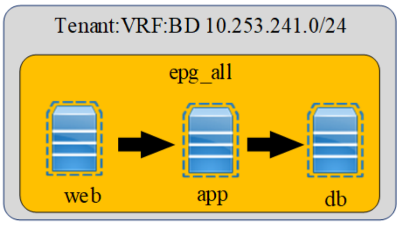

Fig. 7. Experiment with Micro-Segmentation Process

G. Components of The Testbed Simulation

Network infrastructure is a component that consists of hardware and software used to build SDN-based network system testbed simulations based on micro-segmentation-based zero trust security models. Table 1 shows the testbed simulation component consisting of leaf switches, spine switches, and SDN controllers using the Cisco brand. Leaf switches for interconnecting servers to network systems use a 10 GB interface. Spine switch for interconnection between leaf switches using a 40 Gb interface. SDN controller is used to configuring the network system.

TABLE I. THE LIST OF A COMPONENT OF THE CISCO ACI

Name	Bandwidth	Type	Firmware
Spine Switch	40 GB	Nexus 9336PQ	n900-14.2 (3I)
Leaf Switch	10 GB	Nexus 9372PX Nexus 93128TX	n900-14.2 (3I)
SDN Controller	10 GB	APIC Cluster M-1	version 4.2 (3I)

Server infrastructure is a component consisting of hardware and software used to build a testbed simulation system using 3 tier web applications, apps, and DB. The server infrastructure used is based on a virtual machine (VM) under the VMWare brand name. Table 2 shows the testbed simulation component in the form of 3 tier Web, App, and DB.

TABLE II. VIRTUAL MACHINE SPECIFICATION TESTBED

Name	vMemory	vCPU	Operating System
Web	2 GB	1 vCPU	Linux Ubuntu
Application	2 GB	1 vCPU	RHEL 7.8
Database	4 GB	2 vCPU	RHEL 7.8

IV. RESULT AND DISCUSSION

In this paper, the design and implementation of micro-segmentation are built using Cisco ACI. Subsequently, we created three application tiers in the form of a Web, App, and DB in one network segmentation /24. We conducted a testing scenario using a ping mechanism with a variate data transfer to evaluate the performance of the micro-segmentation process. We compared the test results related to the concept of micro-segmentation and without micro-segmentation. Testbed simulation without a micro-segmentation process is done by creating 3 tier web applications, apps, and DB in one EPG with the name epg_all, which contains a Web, App, and DB. Testbed simulation with a micro-segmentation process is done by creating 3 tier Web, App, and DB in 3 EPG, each with epg_web, epg_app, and epg_db.

Testing the performance of testbed simulation with and without a micro-segmentation process is done by sending data packets using the ping mechanism. Data packet delivery is done from the Web to the App server and App server to the DB server, with a large variety of 200 bytes, 500 bytes, 1,000 bytes, 10,000 bytes, and 15,000 bytes many as 100 times the delivery.

Fig. 8 and Fig. 9 show that the average RTT that occurs from the Web to the App in the measurement process without a micro-segmentation process is better than measurement with a micro-segmentation process. Still, the difference in delay is only in the microsecond unit, so it will not affect the network performance significantly. In RTT measurements, the results show that the testbed simulation without running a micro-segmentation process with the lowest average RTT of 285 µs is better than the testbed simulation by running a micro-segmentation process of 289 µs on sending 200-byte and 500-byte data packets. The average addition of 4 µs. These results show that one of the factors that affect the RTT value is the size of the data packet sent and the security policy implemented in the micro-segmentation process.

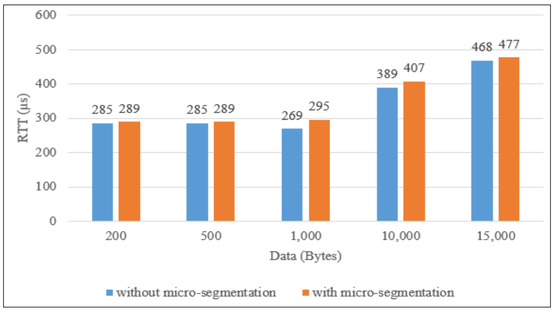

Fig. 8. Average Web RTT measurement results to App

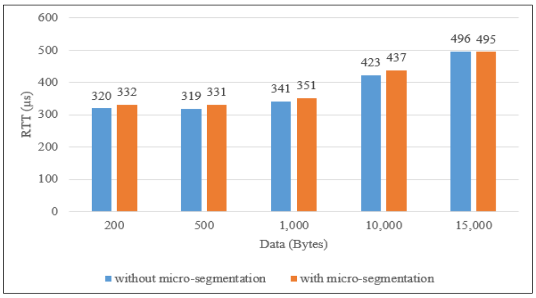

Fig. 9. Average App RTT measurement results to DB

Fig. 10 and Fig. 11 show that the jitter in measurements without a micro-segmentation process is better than the test with a micro-segmentation process. The micro-segmentation process will pass a security policy in the form of a contract (CT) that has been configured on the ACI fabric. In jitter measurement, the results show that the testbed simulation without running a micro-segmentation process with the lowest jitter of 293 µs is better than the testbed simulation by running a micro-segmentation process 304 µs on sending 500-byte data packets. The average addition of jitter is 11 µs.

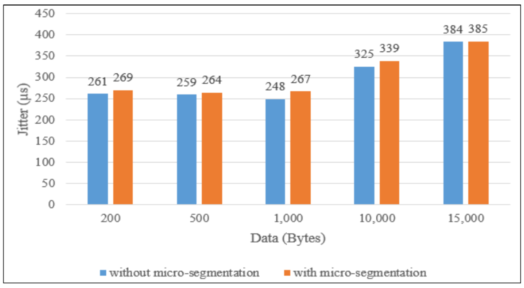

Fig. 10. Jitter measurement results from Web to App

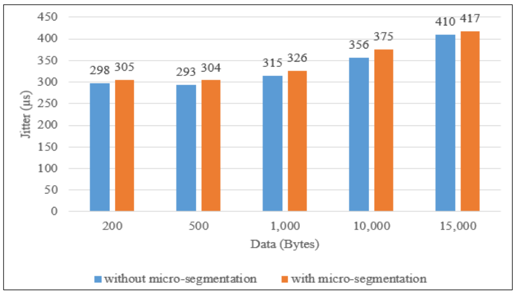

Fig. 11. Jitter measurement results from App to DB

In packet loss measurement, the results show that the testbed simulation without and with the micro-segmentation process succeeded in sending a data packet with no packet loss. These results indicate there is no collision and congestion in network traffic, because the Cisco ACI concept develops as a single switch to the outside world, proficient in bridging and routing. Moving Layer 3 routing to the access layer would restrict the Layer 2 reachability by decoupling Layer 2 domains.

V. Result and Discussion

The results of the analysis of the research have been done by measuring the testbed simulation show that micro-segmentation adds an average round trip time of 4 µs and jitter of 11 µs without packet loss. Based on these results, the security of the data center network can be improved without significantly affecting network performance.

Future work uses a test scenario of high traffic compared to low traffic in the network so that performance comparisons can be obtained. Another approach the performance of a zero trust security model based on micro-segmentation using brands from other vendors such as NSX VMware.

Acknowledgment

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